Strengthening surveillance of and response to foodborne diseases



Introductory manual

World Health Organization

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Strengthening surveillance of and response to foodborne diseases: introductory manual

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WHO also acknowledges with appreciation all experts, institutions, and country representatives who contributed to the set of manuals on strengthening surveillance of and response to foodborne diseases published in 2017. Their

foundational work and insights informed the structure and content of the present manual.

Methods

Approach to development

The working group members engaged in extensive discussions and detailed reviews between January and May 2025, ensuring that the manual reflects emerging challenges and the latest developments in foodborne disease surveillance. The working group met seven times during this period. Their contributions were instrumental in refining the content and shaping this manual into a practical and user-centred resource for countries worldwide.

Decision-making during the development of this manual followed a collaborative and consensus-based process. Draft material prepared by the WHO secretariat was reviewed by the working group members, who provided written and verbal comments through structured consultation rounds. The secretariat considered all inputs and incorporated revisions based on scientific merit, consistency with WHO policies, and practical relevance for country use. Where differing views arose, the secretariat facilitated discussion to reach consensus, and final decisions on content rested with WHO.

The development process was led by the WHO secretariat, which coordinated the drafting, consolidation, and revision of all chapters. The initial draft drew on existing WHO guidance, recent scientific evidence, country experiences, and feedback from users of the 2017 set of manuals on strengthening surveillance of and response to foodborne diseases. Working group members provided technical comments on structure, clarity, and operational relevance, and the secretariat incorporated these inputs through an iterative revision process.

Internal review by WHO technical units ensured coherence across sections and alignment with related WHO materials. This manual includes new information,

clearer explanations, cross-references to related WHO resources, and addresses issues raised by countries using the 2017 set of manuals.

The final manual reflects a collective effort, drawing on expert input, country needs, and WHO standards for accuracy, clarity, and practical application.

Declarations of interest for external contributors

All external contributors involved in preparing this manual, including members of the working group on foodborne disease surveillance, submitted declarations of interest and confidentiality statements to the WHO secretariat. The WHO secretariat reviewed all declarations and found no conflicts of interest sufficient to limit participation in the development of this manual. All contributions were assessed in accordance with WHO policies to ensure that the content reflects independent and impartial technical guidance.

Abbreviations

AI	Artificial intelligence
CDC	Centers for Disease Control and Prevention (of the United States of America)
DALY	disability-adjusted life years
EBS	event-based surveillance
FAO	Food and Agriculture Organization of the United Nations
FBD	Foodborne diseases
GIS	geographic information systems
IBS	indicator-based surveillance
IHR	International Health Regulations
INFOSAN	International Food Safety Authorities Network
JEE	Joint External Evaluation
ORT	Outbreak Response Team
UI	uncertainty interval
WGS	whole genome sequencing
WHO	World Health Organization

What's new in this set of manuals

This set of manuals to strengthen surveillance of and response to foodborne diseases includes several updates and improvements to the set of manuals published in 2017, based on user feedback and advancements in the subject matter. Key changes in each manual include the following:

Introductory manual

- Reference is made to the WHO Global Strategy for Food Safety 2022-2030 *(1)*, the new global foodborne disease estimates to be published by WHO *(2)*, and the Quadripartite guidance on One Health Integrated Surveillance of Antimicrobial Resistance and Antimicrobial Use *(3)*.
- The importance of foodborne disease surveillance and response is revised to include 10 reasons.
- A new table has been included to compare foodborne disease surveillance and food contamination monitoring.
- The guiding principles are revised to include 10 guiding principles.
- The link with the IHR (2005) food safety indicator is explained.
- Several new terms are included in the glossary.
- References are updated accordingly.

Stage one manual (part A): Using indicator- and event-based surveillance to detect foodborne events

- The vision for the surveillance and response system is revised for coherence with the list of strategic goals outlined in the self-assessment.
- Details are included regarding corrective and preventative measures as a function of multisectoral collaboration.
- Decision trees are clearly linked to corresponding strategic goals.
- The STEEEP framework is updated to include a focus on equity.
- References are updated accordingly.

Stage one manual (part B): Investigating foodborne outbreaks

- An explanation of the WHO Outbreak Toolkit *(4)* is provided along with an example of a nationally adapted toolkit *(5)*.
- The questions to help decide if there is an outbreak have been elaborated.
- A structured framework to strengthen hypothesis generation has been introduced.
- References are updated accordingly.

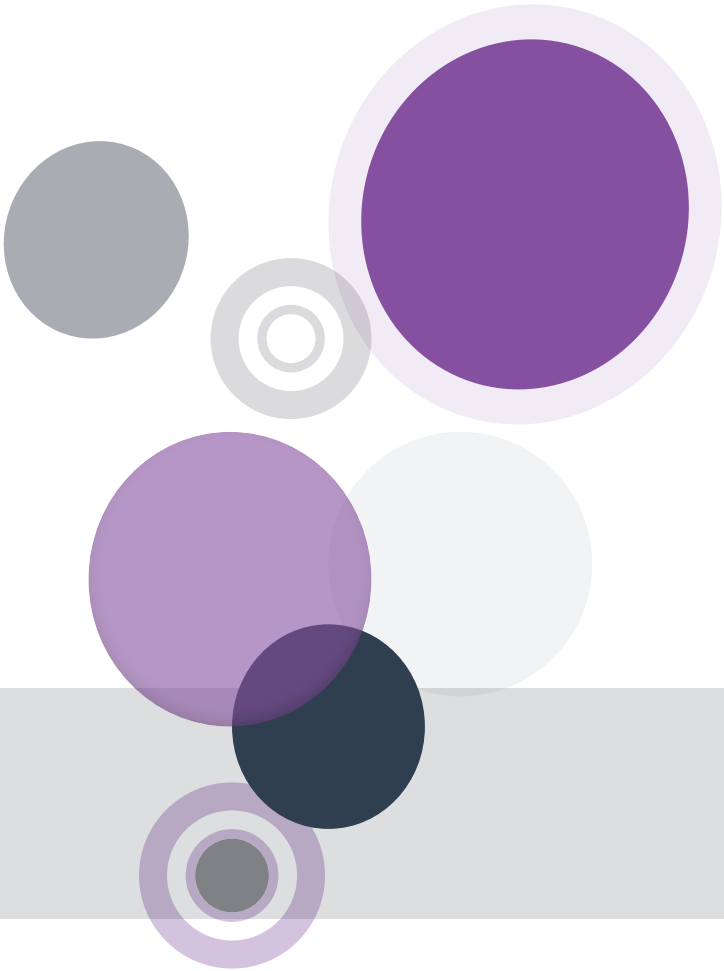
Stage two manual: Strengthening indicator-based surveillance

- The vision for the surveillance and response system is revised for coherence with the list of strategic goals outlined in the self-assessment.
- Decision trees are clearly linked to corresponding strategic goals.
- New decision trees have been added where previously missing (for the sections on rapid risk assessment, response, ad hoc or periodic studies, and monitoring and evaluation).

- New case studies have been included to illustrate concepts discussed.
- Additional web links to surveillance bulletins have been added.
- A new annex has been added to provide an introduction to the potential use of artificial intelligence to support foodborne disease surveillance.
- References are updated accordingly.

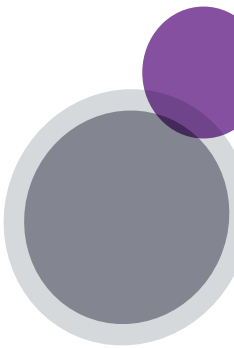
Stage three manual: Integrating surveillance data to better understand risks across the food chain

- Links with the FAO/UNEP/WHO/WOAH One Health Joint Plan of Action are included *(6)*.
- Consideration of climate/environmental health data is discussed.
- Additional examples of integrated surveillance systems for foodborne diseases are included.
- A typology of integration in food chain surveillance is included.
- A discussion on using WHO's Tricycle protocol *(7)* as a foundation for integrated food chain surveillance is discussed and illustrated with a case study, along with guidance on transitioning from a pilot project to a sustainable system.
- A discussion on the use of surveillance data to inform food safety risk management and policy is included.
- A set of example evaluation questions for integrated food chain surveillance systems is provided.
- Common challenges and practical solutions in implementing integrated food chain surveillance are discussed.
- References are updated accordingly.



1

Introduction



1.1 What are foodborne diseases?

Foodborne diseases comprise a wide spectrum of illnesses that result from ingestion of foodstuffs contaminated with microorganisms or chemicals. Food may become contaminated at any stage in the process from production to consumption, and contamination may be the result of environmental contamination, such as pollution of water, soil or air.

Foodborne diseases are a growing public health problem throughout the world and cause a considerable burden of disability and mortality. WHO studies of the global burden of foodborne diseases in 2010 estimated that 31 hazards (including viruses, bacteria, protozoa, helminths and chemicals) caused 600 million foodborne illnesses globally (95% uncertainty interval (UI) 420–960 million) and 420 000 deaths (95% UI 310 000–600 000) (WHO, 2015). The most common clinical presentation of foodborne disease is gastrointestinal symptoms. Other serious consequences include kidney and liver failure, brain and neurological disorders, reactive arthritis and cancer.[1]

1.2 Why are surveillance of and response to foodborne diseases important?

Here are ten reasons why foodborne disease surveillance and response are important:

1. Protect public health

- **Prevent illness and death:** Foodborne diseases are preventable and can be controlled through an effective food safety system that addresses hazards from the place of production to the point of consumption. Surveillance of foodborne diseases can facilitate early

[1] New global estimates to be published by WHO: In 2020, the World Health Assembly, through resolution WHA73.5 mandated WHO to monitor regularly and to report to Member States on the global burden of foodborne and zoonotic diseases at national, regional, and international levels, and in particular to prepare, by 2025, a new report on the global burden of foodborne diseases with up-to-date estimates of global foodborne disease incidence, mortality and disease burden in terms of disability-adjusted life years (DALYs) *(2)*.

detection and intervention, reducing the incidence of illness and preventing potential deaths.

- **Limit and control outbreaks:** Prompt response to identified cases can help contain outbreaks, preventing widespread transmission and protecting public health.

2. Preserve economic interests

- **Minimize negative economic impact:** Foodborne diseases can lead to significant economic losses due to healthcare costs, loss of productivity, and impacts on the food industry. Effective surveillance and response mitigate these financial burdens.
- **Maintain consumer confidence:** Effective surveillance and response maintain consumer trust that authorities are doing their utmost to ensure the safety of food products, which is crucial for the economy.
- **Protect trade:** International food trade has expanded and will continue to grow. Foodborne diseases can impact trade if restrictions are applied to countries identified as sources of contaminated food. Surveillance provides the opportunity to identify and address these issues promptly, thereby safeguarding trade interests.
- **Support travel and tourism:** The travel and tourism industries are particularly vulnerable to health, safety, and environmental concerns, including foodborne diseases. Effective surveillance and response contribute to maintaining the health and safety standards that are crucial for these industries.

3. Understand and address the effects of climate change

- **Recognize emerging threats:** Climate change alters environmental conditions and may accelerate the evolution and emergence of new foodborne pathogens as microorganisms adapt to changing

environmental conditions. Continuous surveillance helps in recognizing these evolving threats.

- **Prevent disease amplification:** Higher temperatures and changing precipitation patterns caused by climate change create ideal conditions for the proliferation of many foodborne pathogens. Surveillance allows for timely responses to these changes, preventing amplified disease transmission.

4. Break the cycle of malnutrition and foodborne diseases

- **Prevent exacerbation of malnutrition:** Malnutrition weakens immune systems, making individuals more susceptible to foodborne diseases. Foodborne diseases can lead to malabsorption of nutrients, further exacerbating malnutrition. Timely interventions help interrupt this cycle, promoting better health outcomes.

5. Contribute to global food security

- **Support safe food systems and protect food supply chains:** Surveillance data helps identify potential hazards in food production, processing, distribution and consumption enabling timely interventions to prevent contamination. Early detection and response to these issues prevent disruptions in food supply chains, thereby supporting global food security and ensuring populations have consistent access to safe and nutritious food.

6. Contribute to global health security

- **Strengthen international collaboration:** Surveillance efforts can foster international cooperation when countries share information, resources, and strategies to combat foodborne diseases collectively.

This collaboration enhances global health security by facilitating a united front against foodborne disease outbreaks. The International Health Regulations (IHR) require countries to notify the WHO of public health events that may be of international concern, including those caused by contaminated food.

- **Prevent cross-border transmission:** By identifying and responding to foodborne diseases promptly, surveillance efforts help prevent the spread of pathogens across borders, contributing to global health security. This includes participating in the FAO/WHO International Food Safety Authorities Network (INFOSAN).

- **Support global health initiatives:** Strengthening surveillance can support global health initiatives and programs, such as the WHO Alliance for Food Safety, aimed at reducing the burden of foodborne diseases, aligning with broader goals of improving global health security.

7. Enhance multisectoral collaboration and the one health approach

- **Facilitate cooperation:** Surveillance and response efforts require collaboration between various sectors, including human, animal, and environmental health sectors, fostering a holistic One Health approach to food safety.

- **Leverage resources:** Collaborative efforts can allow for better resource allocation and utilization, enhancing the effectiveness of foodborne disease control programs.

8. Promote scientific understanding and innovation

- **Foster research:** Continuous monitoring of foodborne diseases and contributes valuable data for scientific research, leading to better

understanding and development of innovative solutions to address them.

- **Improve public health policies:** Data from surveillance can inform public health policies, leading to more effective strategies for foodborne disease prevention and control.

9. Support community health education

- **Raise awareness:** Surveillance programs often include public education components that raise awareness about foodborne diseases and preventive measures.
- **Empower communities:** By educating the public about food safety practices, communities are empowered to take proactive steps in preventing foodborne illnesses.

10. Support foodborne disease burden estimation studies

- **Inform public health priorities:** Surveillance data is crucial for estimating the burden of foodborne diseases, helping to identify the most affected populations and the most common sources of illness. This information is essential for prioritizing public health interventions and resource allocation.
- **Guide policy and funding decisions:** Accurate burden estimates help policymakers and funding agencies make informed decisions about where to focus efforts and invest resources to achieve the greatest impact on reducing foodborne illnesses.

1.3 Purpose of this manual

In line with the WHO Global Strategy for Food Safety 2022-2030 and the International Health Regulations (2005), last revised in 2024, this manual describes

how countries can strengthen their current surveillance and response activities for foodborne diseases, and integrate them in an existing national surveillance and response system. It will enable countries to:

- assess the stage of development of their surveillance and response system in relation to foodborne diseases;
- identify the priorities for developing the surveillance and response system;
- make appropriate decisions about resource allocation for foodborne disease surveillance and response activities;
- facilitate multisectoral collaboration between all the stakeholders involved in disease surveillance and food safety.

1.4 Scope of the manual

All aspects of surveillance, rapid risk assessment, response, investigation, and multisectoral collaboration related to foodborne hazards (microbial, chemical and radiological) that can affect human health are included in this manual.

The following topics are outside the scope of the manual.

- **Strengthening food safety systems.** There are other guidance documents on:
  - strengthening national food control systems *(8,9)*
  - building a national food recall system *(10)*
  - risk-based food inspection *(11)*
  - applying risk analysis principles and procedures during food safety emergencies *(12)*.

- **Using whole genome sequencing as a tool to strengthen foodborne disease surveillance and response.**
  - WHO has published a complementary set of manuals which provides guidance on whole genome sequencing as a tool to strengthen foodborne disease surveillance and response *(13-15)*.

- **Estimating the burden of foodborne diseases.**
  - WHO has published a practical handbook for countries that provides detailed guidance on assessing the burden of diseases caused by microbiological agents commonly transmitted through foods. It is particularly intended for use at national level, and gives a complete picture of the requirements, enabling factors, challenges and opportunities involved, and the steps in the process *(16)*.

- **Chronic diseases related to food.** The causes of these diseases often extend beyond food choices and are monitored through different data collection systems (e.g. hospital admissions, death registers, cancer registries) and population-based surveys. They include:

  - chemical contaminants, such as lead and methylmercury, that can lead to cancer or intellectual disabilities, and prions, which can cause new variant Creutzfeldt–Jakob disease;

  - nutrition-related diseases, including diabetes, cancer and cardiovascular disease.

- **Control and prevention of, and response to, diseases in animals.**

- **Food contamination monitoring.** Food contamination monitoring is an important activity that contributes valuable data to integrated food chain

surveillance systems. Table 1 below provides a comparison between foodborne disease surveillance and food contamination monitoring.

TABLE 1

Comparison of foodborne disease surveillance and food contamination monitoring

Aspect	Foodborne disease surveillance	Food contamination monitoring
Definition	The ongoing, systematic collection, analysis, interpretation, and dissemination of data on foodborne diseases to guide public health action	The systematic sampling and analysis of food products to detect and measure the presence of biological, chemical, or physical hazards
Purpose	To detect outbreaks, assess the burden of foodborne diseases, identify risk factors, and guide public health interventions	To ensure that food products meet safety standards, to detect contamination early, and to prevent unsafe food from reaching consumers
Primary focus	Public health outcomes: cases of illness linked to food consumption	Food products and processes: presence of hazards within the food supply chain
Sectors involved	Public health authorities, epidemiologists, laboratories, hospitals, primary healthcare providers	Food safety authorities, food industry operators, regulatory bodies, laboratories
Data collected	Clinical and epidemiological data (e.g. symptoms, laboratory test results, exposure history), outbreak investigation reports	Analytical data from food sampling (e.g. pathogen counts, toxin levels, chemical residues, physical contaminants), compliance testing results
Approach	Proactive (ongoing surveillance to establish baseline trends and detect emerging threats) and reactive (investigating cases and outbreaks)	Primarily proactive through planned, routine, risk-based monitoring programmes; can also be reactive if targeted testing is done following contamination events, for example
Main outputs	Early detection of outbreaks, estimates of disease burden, identification of at-risk populations and high-risk foods, evaluation of control measures	Verification of compliance with food safety regulations, identification of contaminated foods, risk profiling for hazard management

1.5 Target audience

The primary audience for this manual is public health professionals, such as surveillance and response staff, laboratory staff and food safety staff, who are usually located within the Ministry of Health or health sector.

Other important audiences include:

- competent authorities with a role in food safety, including those responsible for the health of animals and plants or with a role in inspection, trade and commerce;
- consumer groups;
- industries involved in the production and distribution of food;
- development partners, donors and international organizations.

1.6 Guiding principles

The following guiding principles should be kept in mind when working through the steps for strengthening surveillance of and response to foodborne diseases.

1. Foodborne disease surveillance should be integrated into existing national systems

Every country has a national surveillance and response system covering a variety of diseases and syndromes. This manual focuses on building on the existing systems that are part of the core capacity requirements of the IHR, including foodborne diseases, and seeks to avoid the development of a vertical programme for foodborne diseases. While establishing surveillance activities within a vertical programme does allow the surveillance function to remain close to the control function, it can result in the overall surveillance function in a country becoming badly disjointed and inefficient, with field workers participating in multiple complicated systems, using different surveillance methods, terminology and reporting forms and schedules. This entails extra costs and training requirements and can lead to work overload and lack of motivation among health workers.

2. Timely surveillance and response are essential for effective public health management

Surveillance and response are needed in order to respond effectively to acute public health events in a timely manner, so as to minimize the adverse impact on public health, disease burden and the economy. Rapid detection of outbreaks allows for swift intervention measures, such as food recalls, public warnings, and the implementation of control measures to prevent further spread. Delays in response can lead to widespread illness, increased healthcare costs, and significant economic losses due to decreased consumer confidence and international trade restrictions.

3. National surveillance and response systems must adapt to diverse development levels and needs and ensure equity

Surveillance and response systems in countries are at different levels of development and complexity and have different requirements and priorities for future development and capacity-building. Some countries may have well-established systems with advanced laboratory capabilities and robust data management, while others may lack basic infrastructure and trained personnel. Tailoring capacity-building efforts to the specific needs of each country ensures that resources are used efficiently and that improvements in surveillance and response are sustainable and effective. Equity considerations must be integral to these efforts, ensuring that all populations, including vulnerable and marginalized groups, benefit equally from enhanced surveillance and response systems. This involves addressing disparities in healthcare access, resource allocation, and training opportunities across different regions and communities. By prioritizing equity, countries can work towards a more inclusive surveillance and response system that protects the health of all individuals, regardless of socioeconomic status, geographic location, or other factors.

4. External factors drive the enhancement of food safety systems

Developments outside the health sector (e.g. free trade agreements) can also motivate the strengthening of food safety systems. International trade agreements often include stringent food safety standards that require countries to upgrade their surveillance and response mechanisms to comply with these regulations. Additionally, globalization and increased international travel and trade heighten the risk of cross-border transmission of foodborne diseases, underscoring the need for robust food safety systems to protect public health and maintain economic stability.

5. Commitment from senior policy- and decision-makers is crucial for improving foodborne disease surveillance

It is vital that senior policy- and decision-makers are committed to improving surveillance and response to address foodborne diseases. High-level support ensures that adequate funding, legislative backing, and intersectoral collaboration are provided. Decision-makers play a crucial role in prioritizing food safety within national health agendas, mobilizing resources, and fostering a culture of continuous improvement and accountability within surveillance and response systems.

6. Adequate human and financial resources are necessary for system development

Sufficient resources, both human and financial, are required to support the development of the system. Investment in training programs for health workers, laboratory technicians, and epidemiologists is essential to build the necessary skills and expertise. Financial resources are needed to upgrade laboratory equipment, develop and maintain data management systems, and conduct routine monitoring and evaluation of surveillance activities. Without adequate resources, efforts to strengthen surveillance and response systems may be hampered, leading to gaps in detection and delayed responses to foodborne disease outbreaks.

7. International networks should be leveraged to enhance national surveillance and response systems

Countries should draw on the resources of existing international networks to strengthen national surveillance and response systems. Organizations such as the WHO provide technical assistance, training, and access to best practices and guidelines. Participation in international networks facilitates information sharing, early warning of potential threats, and coordinated response efforts, which are critical for managing foodborne diseases that may spread across borders. Leveraging these resources helps countries build resilient and interconnected surveillance systems.

8. Multisectoral collaboration is essential for comprehensive food safety

Effective surveillance and response to foodborne diseases require collaboration across multiple sectors, including health, agriculture, the food industry, and regulatory authorities. Establishing strong intersectoral partnerships ensures that all relevant stakeholders of the food system are engaged in the process, from monitoring food production and distribution to implementing control measures during outbreaks. Collaborative efforts enhance information sharing, resource allocation, and coordinated action, which are essential for a holistic approach to food safety.

9. Technology and innovation should be utilized to enhance surveillance capabilities

Modern technology and innovative solutions can significantly improve the efficiency and effectiveness of foodborne disease surveillance systems. Implementing digital tools for real-time data collection, analysis, and reporting enables faster detection and response to outbreaks. Technologies such as whole genome sequencing (WGS), geographic information systems (GIS), mobile applications, artificial intelligence (AI) and big data analytics have the potential to enhance surveillance and response activities, facilitate communication among stakeholders, and support evidence-based decision-making.

10. International cooperation and information exchange are crucial for effective surveillance and response

Foodborne diseases do not respect national borders, making international cooperation and information exchange crucial for effective surveillance and response. Countries should actively participate in global and regional initiatives, share data on foodborne disease outbreaks, and collaborate on research and capacity-building efforts. Strengthening ties with international organizations and neighbouring countries helps to coordinate responses to transboundary threats, harmonize surveillance standards, and enhance the overall resilience of food safety systems.

1.7 Terminology

Some of the terms used in this manual can have different meanings in different settings. A glossary of terms, as they are used here, is given in Annex 1. The use of these terms is discussed in more detail in Annex 2, including the different approaches to risk in the different sectors with a stake in foodborne diseases.

Throughout the manual, the term "case" has been used to refer to any person who meets a case definition, either for surveillance purposes or during an outbreak investigation. The term case is used because:

- case is a standard way to describe a person counted as part of an outbreak investigation; a case can be further identified as a suspect case, a probable case or a confirmed case;
- not all suspect and confirmed cases are patients (patient refers to someone who receives medical care or is registered with a medical service);
- ill people or persons does not cover all cases: in some situations asymptomatic cases may be included, as well as people who are well at the time of the investigation but had symptoms in the past;
- case-finding and case definition are standard epidemiological terms.

2

Framework for strengthening surveillance of and response to foodborne diseases

The approach to strengthening the surveillance of and response to foodborne diseases builds on the systems countries are required to develop under the IHR. Such systems have four key components: indicator-based surveillance, event-based surveillance, rapid risk assessment and response. For foodborne diseases, additional targeted ad hoc studies may be needed to answer specific questions that surveillance and response data alone cannot answer. In addition, the control of foodborne diseases requires strong multisectoral collaboration, involving the surveillance and response, food safety, and agriculture sectors.

Definitions of these various components, together with a discussion of their role in strengthening surveillance and response for foodborne diseases, are given below.

1

Indicator-based surveillance (IBS)

The regular systematic collection, monitoring, analysis and interpretation of structured data related to a case or syndrome definition *(17)*. Data about illness in individuals (either syndromes or laboratory results) should be recorded systematically, analysed, interpreted and disseminated. Thresholds can be applied to the data to detect outbreaks and the data can be used to monitor trends and evaluate interventions.

2

Event-based surveillance (EBS)

The organized collection, monitoring, assessment and interpretation of unstructured information about health events that may represent a risk to public health *(17)*. Rumours or reports of foodborne events that pose a threat to public health may come through official networks or from informal sources, such as the media or the community.

3

Rapid risk assessment of acute public health events

A systematic process for gathering, assessing and documenting information about acute public health events with the aim of assigning a level of risk *(18)*. It provides the basis for action to manage and reduce the negative consequences of acute public health risks. In the context of this manual, an acute public health event is any outbreak or rapidly evolving situation that may have negative consequences for human health and requires immediate assessment and action. The term includes events that have not yet led to disease in humans but have the potential to cause disease through exposure to infected or contaminated food, water, animals, manufactured products or environments.

4

Response

Action taken once a foodborne event has been detected (using EBS or IBS) to rapidly investigate the event, identify its causes, prevent further spread and strengthen future prevention efforts.

5

Ad hoc or periodic studies

Targeted public health studies designed to answer specific questions relating to foodborne diseases, e.g. total diet studies, burden of disease studies, source attribution studies.

6

Multisectoral collaboration

Staff from surveillance and response, food safety, animal health, environmental health and other relevant sectors working together to reduce the burden of foodborne diseases in the community.

Most multisectoral collaboration occurs within risk-based processes designed to understand why there is a foodborne problem, assess the extent of the problem and manage the risks to keep future illness to a minimum. The main risk-based processes of relevance to foodborne diseases are the rapid risk assessment of acute foodborne events, risk profiling and risk analysis. These risk-based processes are described in more detail in Annex 2.

To support countries in strengthening the surveillance and response system for foodborne diseases, this manual describes three stages of development. Each stage builds on the previous one in a stepwise manner, as shown in Fig. 1. Table 2 outlines how the stages evolve as the system is progressively strengthened. A country does not need to have developed all of the components of one stage before moving on to the next.

1.1 Strategic goals are outlined for each of the three stages:

Stage 1

- ☑ Strategic goal 1. An indicator-based surveillance system that can monitor trends of disease syndromes and identify outbreaks of foodborne diseases;
- ☑ Strategic goal 2. An event-based surveillance system capable of detecting foodborne events;
- ☑ Strategic goal 3. Capacity to undertake rapid risk assessments of acute public health events such as foodborne disease outbreaks;
- ☑ Strategic goal 4. Outbreak response teams (at national level, as a minimum) can gather epidemiological evidence during foodborne outbreaks;
- ☑ Strategic goal 5. Outbreak response teams have the capacity to collect and transport appropriate specimens to a laboratory for identification of the etiological agent during foodborne outbreaks;

- ☑ Strategic goal 6. Multisectoral collaboration that facilitates rapid information exchange and support during foodborne disease outbreaks;
- ☑ Strategic goal 7. Monitoring and evaluation of the surveillance and response system for foodborne diseases.

Stage 2

- ☑ Strategic goal 8. An indicator-based surveillance system that includes laboratory analysis, to allow better understanding of trends in foodborne diseases and to increase the sensitivity and specificity of outbreak detection;
- ☑ Strategic goal 9. There is a fully functional notifiable disease surveillance system that can successfully monitor trends and detect outbreaks of foodborne diseases;
- ☑ Strategic goal 10. A fully functional event-based surveillance system capable of detecting foodborne events;
- ☑ Strategic goal 11. The capacity for rapid risk assessment of foodborne events is strengthened at the subnational level;
- ☑ Strategic goal 12. Capacity exists at subnational level to carry out analytical epidemiological studies during foodborne disease outbreak investigations;
- ☑ Strategic goal 13. Ad hoc or periodic research studies are conducted to gain a better understanding of foodborne diseases and their sources;
- ☑ Strategic goal 14. Multisectoral collaboration facilitates the sharing of data for risk profiling;
- ☑ Strategic goal 15. Monitoring and evaluation of the surveillance and response system for foodborne diseases.

Stage 3

- [x] Strategic goal 16. Integrated food chain surveillance system that allows a better understanding of risks across the food chain.

FIG. 1

Stepwise approach to strengthening surveillance of and response to foodborne diseases

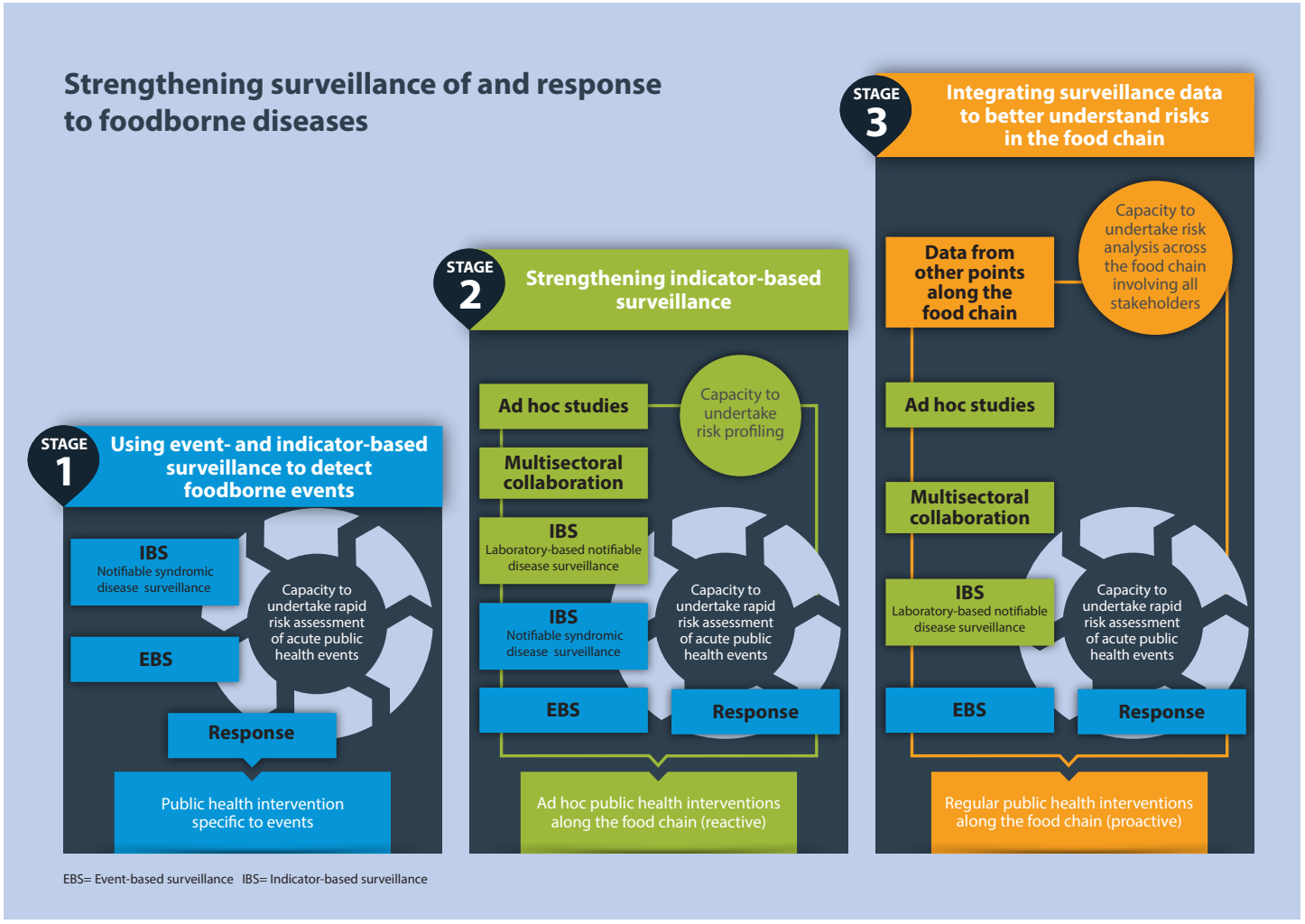


TABLE 2

The three stages of strengthening surveillance of and response to foodborne diseases

	Stage 1	Stage 2	Stage 3
Key components of the surveillance and response system	• IBS: notifiable conditions (syndromes/diseases) • EBS • Rapid risk assessment • Response • Multisectoral collaboration • Public health interventions for specific foodborne disease events	• IBS: notifiable diseases • IBS: laboratory-based surveillance • EBS • Rapid risk assessment • Response • Targeted ad hoc studies • Multisectoral collaboration • Ad hoc public health interventions along the food chain	• IBS: notifiable diseases • IBS: laboratory-based surveillance • EBS • Rapid risk assessment • Response • Targeted ad hoc studies • Multisectoral collaboration • Integrating surveillance data from points along the food chain • Regular public health interventions along the food chain
Focus of development at each stage			
Surveillance and response	Strengthen: • IBS to detect outbreaks of priority foodborne diseases • EBS to detect foodborne events, including outbreaks • Databases to ensure they capture relevant data from the IBS and EBS • epidemiological evidence gathered during investigations of foodborne outbreaks • laboratory evidence gathered during investigations of foodborne outbreaks	Strengthen: • IBS to include laboratory analysis for priority foodborne diseases • notifiable disease surveillance system to detect outbreaks and monitor trends • Databases to ensure that they can capture date from the expanding IBS and EBS systems • sensitivity and specificity of EBS to detect more foodborne events • subnational response capacity to carry out analytical epidemiology during foodborne outbreak investigations • capacity to undertake targeted ad hoc studies	• Surveillance and response are fully functional, but mechanisms need to be created for sharing data on foodborne disease with other sectors across the food chain • A database to house the data coming from across the food chain

(TABLE 2 Cont.)

	Stage 1	Stage 2	Stage 3
Risk-based processes	Rapid risk assessment of acute public health events	• Rapid risk assessment of acute public health events • Risk profiling	• Rapid risk assessment of acute public health events • Risk analysis across the food chain
Identifying priority diseases for surveillance	Desktop review, based on available data	Strategy grids and/or Delphi panel	Multisectoral consideration of available data from across the food chain that can be shared
Multisectoral collaboration	Operational links established between surveillance and response, food safety, animal health and laboratories	Surveillance and response, food safety, laboratories and agriculture staff work together to consider the risks and interventions	Multisectoral collaboration is at its peak, with all sectors regularly considering data and risks across the food chain
Public health interventions to minimize the impact of foodborne diseases	Interventions occur at the local level in response to events, including foodborne outbreaks	• Interventions occur at the local level in response to events, including foodborne outbreaks • Interventions are reactive, based on risk profiles • Interventions occur through national policy (e.g. food standards and regulations)	• Interventions occur at the local level in response to events, including foodborne outbreaks • Ongoing interventions are proactive across the food chain based on risk analysis • Interventions occur through national policy (e.g. food standards and regulations)
Sectors responsible	Mainly surveillance and response sector	• Surveillance and response for strengthening IBS • Risk profiling involves the surveillance and response, food safety, animal health and other relevant sectors • Interventions are largely the mandate of the agriculture and food safety sectors	• Surveillance and response system provides data from illness in humans to the integrated food chain surveillance system • Interventions are largely the mandate of agriculture and food safety sectors

1.2 Link with the IHR (2005) food safety indicator

Having a surveillance system in place for the detection and monitoring of foodborne diseases and food contamination is one of the food safety capacity indicators under the IHR (2005). It measures the capacity to detect food safety events and is monitored through the Joint External Evaluation (JEE) process. It is scored categorically from 1 to 5 as per Table 3 below *(19)*.

TABLE 3

JEE capacity assessment for surveillance of foodborne diseases and contamination

Level 1	No or very limited surveillance system in place for foodborne diseases or food contamination (chemical and microbiological) monitoring
Level 2	Country has IBS or EBS and monitoring systems in place to monitor and detect foodborne events (outbreak or contamination)
Level 3	IBS or EBS system includes laboratory analysis to assign etiology of foodborne diseases or origin of contamination event, and investigate hazards in foods linked to cases, outbreaks or events
Level 4	ountry has capacity to undertake rapid risk assessments of acute foodborne events at the national and intermediate levels
Level 5	Country has a surveillance system in place that integrates information from the entire food chain including timely and systematic information exchange, to enable a better understanding of risk and mitigation possibilities

Working through the stage one manual (part A) will help countries to achieve Level 2 (limited capacity); working through the stage one manual (part B) and the stage two manual will help countries to achieve Level 3 (developed capacity) or 4 (demonstrated capacity); and working through the stage three manual will help countries to achieve Level 5 (sustainable capacity).

These scores are being regularly monitored as part of the assessment of the progress indicator on foodborne disease surveillance as described in the WHO Global Strategy for Food Safety 2022-2030 *(1)*.

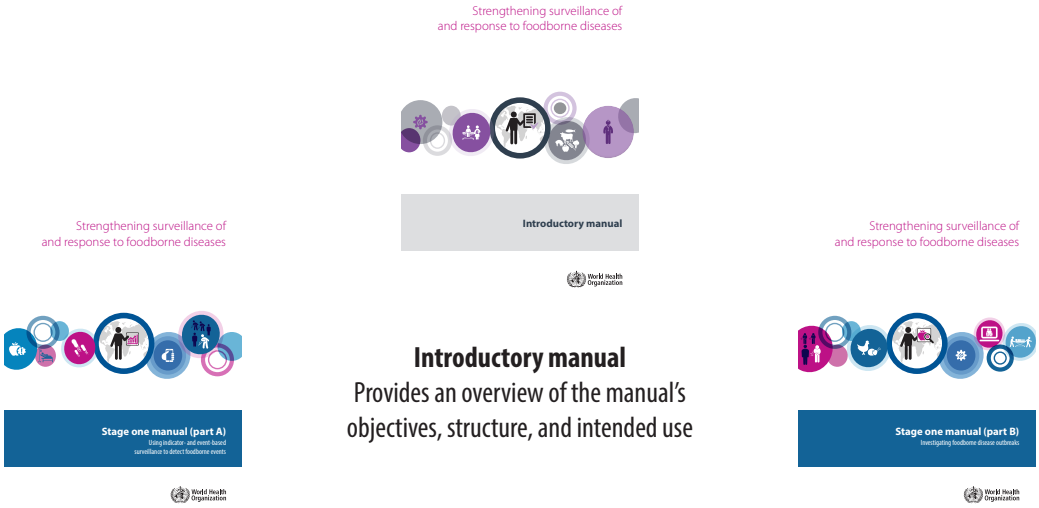
3

Structure of this set of manuals

Across five manuals, users are guided through key stages of strengthening FBD surveillance and response systems. To ensure clarity and ease of reference, each manual is assigned a title and a corresponding number. The numbering system reflects the progression across the three stages while distinguishing between the two manuals under stage 1, as per Fig. 2.

FIG. 2

Overview of the manuals



Introductory manual
Provides an overview of the manual's objectives, structure, and intended use

Stage one manual (part A): Using indicator- and event-based surveillance to detect foodborne events
Focuses on detecting foodborne events using both indicator-based and event-based surveillance approaches

Stage one manual (part B): Investigating foodborne disease outbreaks
Guides users on the investigation of foodborne disease outbreaks, including outbreak detection and response

Stage two manual: Strengthening indicator-based surveillance
Outlines strategies to enhance indicator-based surveillance systems for foodborne diseases

Stage three manual: Integrating surveillance data to better understand risks across the food chain
Discusses the integration of data from multiple surveillance sources to understand and address risks across the food chain

Each stage includes the following information.

Minimum requirements

The capacities required for a country to enter the stage being described.

Objectives

The objectives of the surveillance and response system for foodborne diseases at the stage being described.

Vision

A statement about how the surveillance and response system should perform when all the capacities needed for the stage are in place. A list of the strategic goals to be achieved to obtain the vision in this stage is also included. A diagram shows the capacities to be strengthened in the stage.

Components and risk-based processes

A description of the key components and the risk-based processes: the text sets out a series of logical steps to be taken to develop capacities in each component and process, and includes tools for use by countries.

Case studies

These are used throughout to provide examples of actions taken in different countries to strengthen surveillance and response for foodborne diseases.

Managing implementation

A template allows countries to document their current capacities and helps identify priorities for strengthening surveillance and response to foodborne diseases. The completed template could be attached to existing strategic plans for strengthening the national surveillance and response system.

Monitoring and evaluation

Examples are provided of monitoring indicators and attributes that can be evaluated to assess the performance of the surveillance and response system in relation to foodborne diseases.

1.1 How to use this manual

The steps involved in using this manual are shown in Fig. 3.

Step 1. Self-assessment

Each country needs to start by assessing the stage of development of the surveillance and response system for foodborne diseases. Some components of the system may be in stage 1 while others may be in stage 2. For example, the notifiable disease surveillance system may include some laboratory-confirmed diseases (stage 2), but there may be limited capacity to conduct analytical epidemiology as part of the response to foodborne disease outbreaks (stage 1).

Annex 3 contains a self-assessment tool, with a list of indicators describing the capacities required to conduct the surveillance of and response to foodborne diseases in each of the three stages. The tool can be used to assess the capacities that currently exist in a country and to identify areas where further development is required.

Step 2. Identifying the relevant tools and guidance

The structure of the country self-assessment tool in Annex 3 matches the structure of the manuals. Each of the strategic goals and indicators is dealt with in the manuals.

If the self-assessment shows that a country does not possess the necessary capacities for a particular activity, the manual provides specific advice on how to build capacity.

Step 3. Using the decision-trees

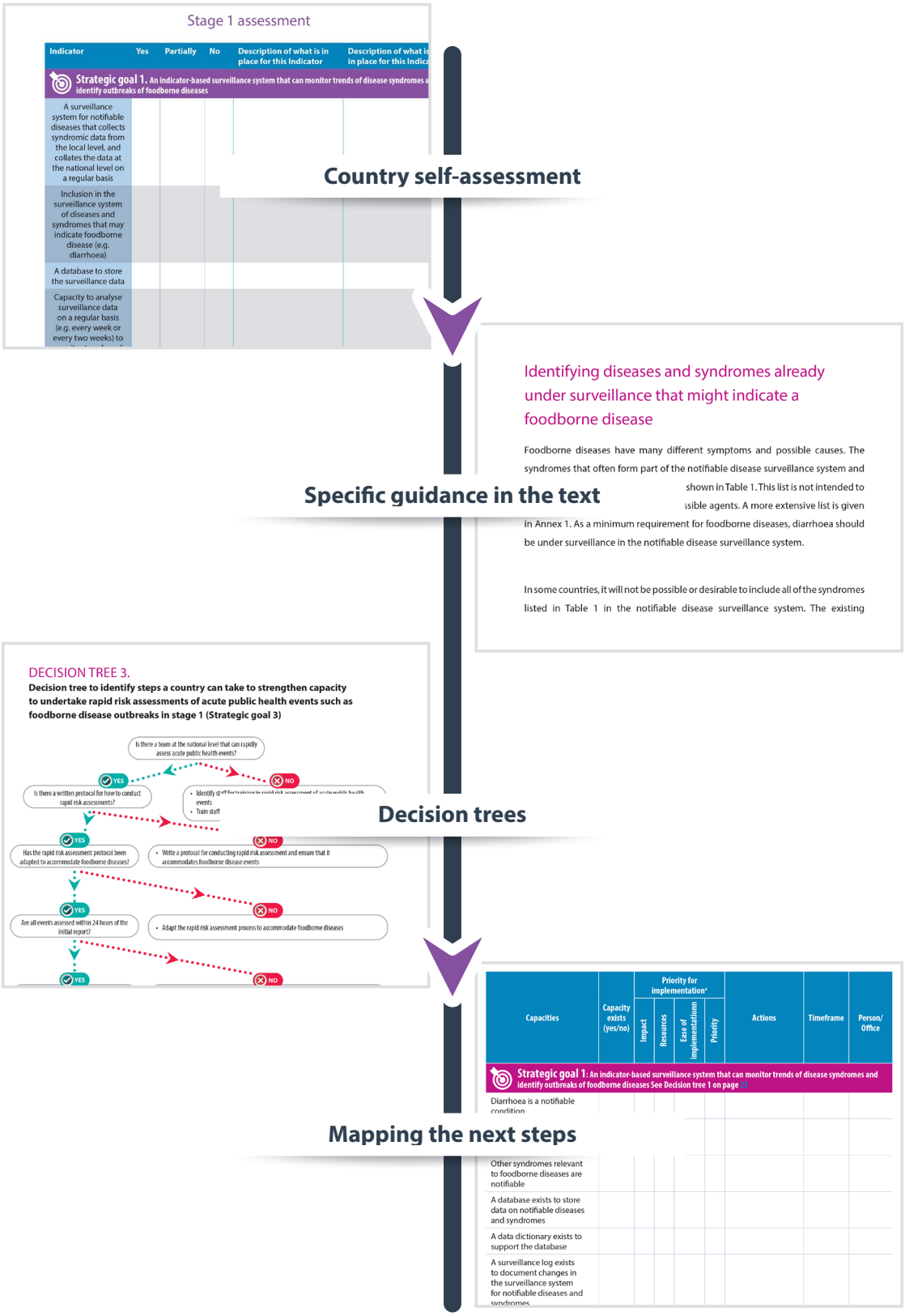
In most of the sections of the manual, a decision-tree provides a step-by-step guide to the capacities required for logical and sustainable development of the surveillance and response system in relation to foodborne diseases. Some of the branches in the decision-trees correspond to rows in the self-assessment tool. Once the self-assessment is completed, a country can refer to the decision-trees and the relevant sections in the manual to determine the steps needed to strengthen the surveillance and response system. Accompanying the decision-trees is specific guidance in the text or supporting tools that countries can use in strengthening activities.

Step 4. Mapping the next steps

The "Managing implementation" section of each stage manual contains a tool that can help countries map the next steps needed to develop their system. Each branch of the relevant decision-tree corresponds to a row in the template table for managing implementation. Countries can identify specific areas to focus on by prioritizing actions. It is not necessary to follow up on every identified action. Through the prioritization process, country can identify specific tasks that can be completed within a 2–3 year timeframe, as well as look at some of the capacities that need to be developed in the longer term.

FIG. 3

How to use this set of manuals



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Annex 1

Glossary of terms used in this manual

The definitions provided here apply to the terms as used in the five manuals and may differ from those given in other documents.

Active surveillance.

Surveillance initiated by a health authority to search systematically for cases of disease or events. It includes reaching out to health care workers, laboratories and members of the community to actively search for cases of specific diseases or syndromes.

Acute public health event.

Any event that represents an immediate threat to human health and requires prompt action, i.e. implementation of control or mitigation measures to protect the health of the public. It includes events that have not yet led to disease in humans but that have the potential to cause such disease through exposure to infected or contaminated food, water, animals, manufactured products or environments, or as a result of direct or indirect consequences of natural events, conflicts or other disruptions of critical infrastructure *(3)*.

Ad hoc studies.

Targeted public health studies designed to answer specific questions relating to foodborne diseases, e.g. total diet studies, burden of disease studies, source attribution studies.

Agent.

A factor, e.g. microorganism, chemical substance or radiation, whose presence or excessive presence is essential for the occurrence of disease.

Artificial Intelligence.

Artificial intelligence generally refers to the performance by computer programs of tasks that are commonly associated with intelligent beings. The basis of AI is algorithms, which are translated into computer code that carries instructions for rapid analysis and transformation of data into conclusions, information or other outputs *(5)*.

All-hazards approach.

An approach that takes into consideration all possible hazards, including biological, chemical, and radionuclear hazards *(3)*.

Analytical epidemiology.

The study of hypothesized causal relationships to make causal inferences. An analytical study is usually concerned with identifying or measuring the effects of risk factors or the health effects of specific exposures or interventions. This is in contrast to descriptive studies, which do not test causal hypotheses. Cohort studies and case–control studies are examples of analytical epidemiological studies *(6)*.

Big data.

Big data are complex data that are rapidly collected in such unprecedented quantities that terabytes (one trillion units [bytes] of digital information), petabytes (1000 terabytes) or even zettabytes (one million petabytes) of storage space may be required as well as unconventional methods for their handling. The unique properties of big data are defined by four dimensions: volume, velocity, veracity and variety *(5)*.

Case.

Any person who meets a case definition, either for surveillance purposes or during an outbreak investigation.

Case definition.

A set of criteria (not necessarily diagnostic) that must be fulfilled in order to identify a person as having a particular disease or condition. Case definitions can be based on geographical, clinical, laboratory, or combined clinical and laboratory criteria *(6)*.

Capacity.

The ability of individuals, institutions and societies to perform functions, solve problems, and set and achieve objectives in a sustainable manner *(7)*.

Cluster-based surveillance.

A type of indicator-based surveillance in which specimens are collected from patients in whom clinicians have recognized a clustering of a particular clinical syndrome, by time, place or person.

Collaborative surveillance.

The systematic strengthening of capacity and collaboration among diverse stakeholders, both within and beyond the health sector, with the ultimate goal of enhancing public health intelligence and improving evidence for decision-making *(8)*.

Competent authority.

The regulatory authorities designated under the food control system that help ensure food safety and quality along the food chain and manage food safety hazards, fraud issues, emerging risks and emergencies.

Context assessment.

Assessment of the environment in which an event takes place *(4)*.

Delphi method.

Iterative circulation to a panel of experts of questions and responses; the questions are progressively refined in light of the responses to each round of questions. The aim is to arrive at a consensus on an issue or problem without allowing any one participant to dominate the process *(6)*.

Descriptive epidemiology.

The organization and summarization of health-related data according to time, place and person characteristics *(9)*. A case series investigation is an example of a descriptive epidemiological study.

Desktop review.

An assessment of existing documentation and published scientific literature in order to make informed decisions about priorities for the surveillance and response system.

Exposure assessment.

Evaluation of the potential exposure of individuals and populations to hazards identified in a hazard assessment *(4)*.

Evaluation.

The periodic assessment of the relevance, effectiveness and impact of activities in relation to the objectives of the surveillance and response system *(10)*.

Event.

A manifestation of disease or an occurrence that creates a potential for disease. Events may be related to infections, zoonoses, breaches of food safety, or chemical, radiological or nuclear contamination, and transmission may be from person to person or via vectors, animals, goods, food or the environment *(3)*.

Event-based surveillance.

The organized collection, monitoring, assessment and interpretation of unstructured information about health events that may represent a risk to public health *(3)*.

Field Epidemiology Training Programme (FETP).

FETPs are two-year work-based training programmes designed to increase epidemiological and public health capacity. Trainees are based in a public health organization, such as a ministry of health. During their placements, the trainees acquire the knowledge, skills and competencies required to be an epidemiologist and public health practitioner through learning-by-doing.

Focal point.

A person who is nominated as the point of contact for an activity or process.

Food.

Any substance, whether processed, semi-processed or raw, that is intended for human consumption; it includes drink, chewing gum and any substance used in the manufacture, preparation or treatment of food, but does not include cosmetics, tobacco or substances used only as drugs *(11)*.

Food chain.

The series of processes that food goes through; it includes primary production (including feeds, agricultural practices and environmental conditions), product design and processing, transport, storage, distribution, marketing, preparation and consumption *(12)*.

Foodborne disease.

Any disease of an infectious or toxic nature caused by the consumption of food *(9)*.

Foodborne event.

Any event related to the occurrence of disease in humans that is caused by contaminated food (e.g. an outbreak of salmonellosis caused by improperly handled eggs) or that has the potential to expose humans to known or suspected hazards through food (e.g. accidental or intentional contamination of food with chemicals).

Foodborne disease outbreak.

For common diseases (such as salmonellosis), the occurrence of two or more cases resulting from ingestion of the same food. For rare diseases, e.g. botulism, one case may be considered an outbreak *(9)*.

Food safety.

Actions taken to ensure that food will not cause harm to the consumer when it is prepared and eaten according to its intended use *(1)*.

Food System.

Food systems embrace the entire range of actors and their interlinked value-adding activities involved in the production, aggregation, processing, distribution, consumption, and disposal (loss or waste) of food products that originate from agriculture (including livestock), forestry, fisheries, and food industries, and the broader economic, societal, and natural environments in which they are embedded (Scientific Group of the UN Food Systems Summit, 2021)

Hazard.

An agent that has the potential to cause adverse health effects in exposed populations *(4)*.

Hazard assessment.

Identification of the hazard (or potential hazards) causing a foodborne disease event and of the associated adverse health effects *(4)*.

Indicator-based surveillance.

The regular, systematic collection, monitoring, analysis and interpretation of structured data, i.e. of indicators produced by a number of well identified, mostly health-based, formal sources *(3)*.

Integrated food chain surveillance.

The routine sharing of data and information between the public health, food safety and animal health sectors in order to direct control measures to minimize the burden of foodborne diseases.

Laboratory-based surveillance.

A form of indicator-based surveillance of cases that have been confirmed by a laboratory test. The laboratories that perform the testing report the results to the surveillance system, as well as informing the clinicians who requested the tests.

Machine learning.

The various types of AI technology include machine-learning applications such as pattern recognition, natural language processing, signal processing and expert systems. Machine learning, which is a subset of AI techniques, is based on use of statistical and mathematical modelling techniques to define and analyse data. Such learned patterns are then applied to perform or guide certain tasks and make predictions *(5)*.

Modified Field Epidemiology Training Programme.

A shortened version of the Field Epidemiology Training Programme (i.e. of less than 2 years' duration), which targets basic knowledge and competencies in surveillance and outbreak investigation that are specifically designed for resource-limited settings.

Monitoring.

The routine and continuous tracking of the implementation of planned surveillance activities and of the overall performance of the surveillance and response system *(10)*.

Multisectoral collaboration.

Multiple sectors working together to achieve common objectives, goals and tasks with shared responsibility. In the context of foodborne diseases, this will include staff from the public health surveillance and response sector, food safety sector, animal health sector, environmental health and other relevant sectors.

Notifiable disease.

A disease that, because of its public health importance, must be reported to the public health authority under legislation or decree, in the pertinent jurisdiction when a diagnosis is made *(6)*.

One health.

One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals and ecosystems. It recognizes the health of humans, domestic and wild animals, plants, and the wider environment (including ecosystems) are closely linked and inter-dependent. The approach mobilizes multiple sectors, disciplines and communities at varying levels of society to work together to foster well-being and tackle threats to health and ecosystems, while addressing the collective need for clean water, energy and air, safe and nutritious food, taking action on climate change, and contributing to sustainable development *(13)*.

Outbreak Response Team (ORT).

A team of people tasked with investigating and controlling the outbreak.

Passive surveillance.

Surveillance for diseases or events that is initiated by the source of the data, such as a health care provider or laboratory.

Public health surveillance.

The systematic continuous collection, collation and analysis of data for public health purposes and the timely dissemination of public health information for assessment and public health response as necessary *(9)*.

Rapid response team.

The team responsible for investigating public health events.

Annex 2 gives more detailed definitions of risk-related terms and describes how the definitions differ between the disciplines of food safety and surveillance and response.

Response.

Any public health action (e.g. event monitoring, providing information to the public, field investigations and control or mitigation measures) triggered by the detection of a public health risk *(3)*.

Risk.

The likelihood that an adverse event will occur during a specified period and the likely magnitude of its consequences *(4)*.

Rapid risk assessment.

A systematic process for gathering, assessing and documenting information to assign to an event a level of risk to human health. Rapid risk assessment includes three components: hazard assessment, exposure assessment and context assessment. The rapid risk assessment provides the basis for deciding on action to be taken to manage and reduce the negative consequences of acute public health events *(3)*.

Sensitivity.

The proportion of actual cases in a population that is detected and reported through the surveillance system *(10)*.

Sentinel surveillance.

Surveillance based on a selected population, with samples chosen to represent the relevant experience of particular groups. Standard case definitions and

protocols are used to ensure that comparisons across time and sites are valid *(6)*. Examples of sentinel surveillance include data collected from a subset of doctors or laboratories, or from sentinel geographical locations.

Severity-based surveillance.

A type of indicator-based surveillance in which specimens are collected from patients with severe illness (e.g. those admitted to hospital) or who have died from a suspected foodborne illness.

Specificity.

The proportion of persons without the disease in a population that are considered by the surveillance system as not having the disease. A surveillance system with low specificity would pick up false outbreaks, resulting in a waste of resources for their investigation *(10)*.

Strategy grid.

A 2 x 2 grid that shows the relationship between two criteria being evaluated. It can be used when deciding on priorities for surveillance and response.

Surveillance and response system.

The existing infrastructure, staff and processes used for surveillance of and response to communicable diseases.

Surveillance and response for foodborne diseases.

Use of existing surveillance and response systems for foodborne diseases.

Syndrome.

A group of clinical signs and symptoms that consistently occur together, or a condition characterized by a set of associated clinical signs and symptoms *(4)*.

Syndromic surveillance.

A method of surveillance that uses health-related data based on clinical observations rather than laboratory confirmation of diagnoses *(14)*.

Timeliness.

Refers to an acceptable length of time between steps in a public health surveillance system. For example, the time between onset of illness and specimen collection or between laboratory confirmation of disease and notification needs to be within certain limits.

Whole genome sequencing.

A laboratory technique that can be used to determine the complete nucleic acid sequence of an organism's genome (i.e. the order and number of the nucleotides, adenine (A), guanine (G), cytosine (C) and thymine (T), which make up an organism's genetic code) *(2)*.

Vertical programme.

A programme in which interventions are provided through delivery systems that typically have separate administration and budgets, with different levels of structural, funding and operational integration in the wider health system.

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Annex 2

Explanation of risk-related terms used in this manual

A number of key risk-related terms are used throughout this manual. This annex discusses the terms used and describes how risk is considered differently in the different sectors with a stake in foodborne diseases.

How risk is considered differently between sectors

Fig. A2.1 describes how the emphasis on risk-related terms changes as the surveillance and response system evolves. In stage 1, the focus is mainly on detecting events and the risk terminology is largely in the realm of the surveillance and response sector. In stage 3, with integrated food chain surveillance, the terminology is focused on the Codex Alimentarius risk definitions *(1)*. A comparison of risk-related definitions is given in Table A2.1.

FIG. A2.1

How risk-related terms are used in each of the three stages of surveillance of and response to foodborne diseases

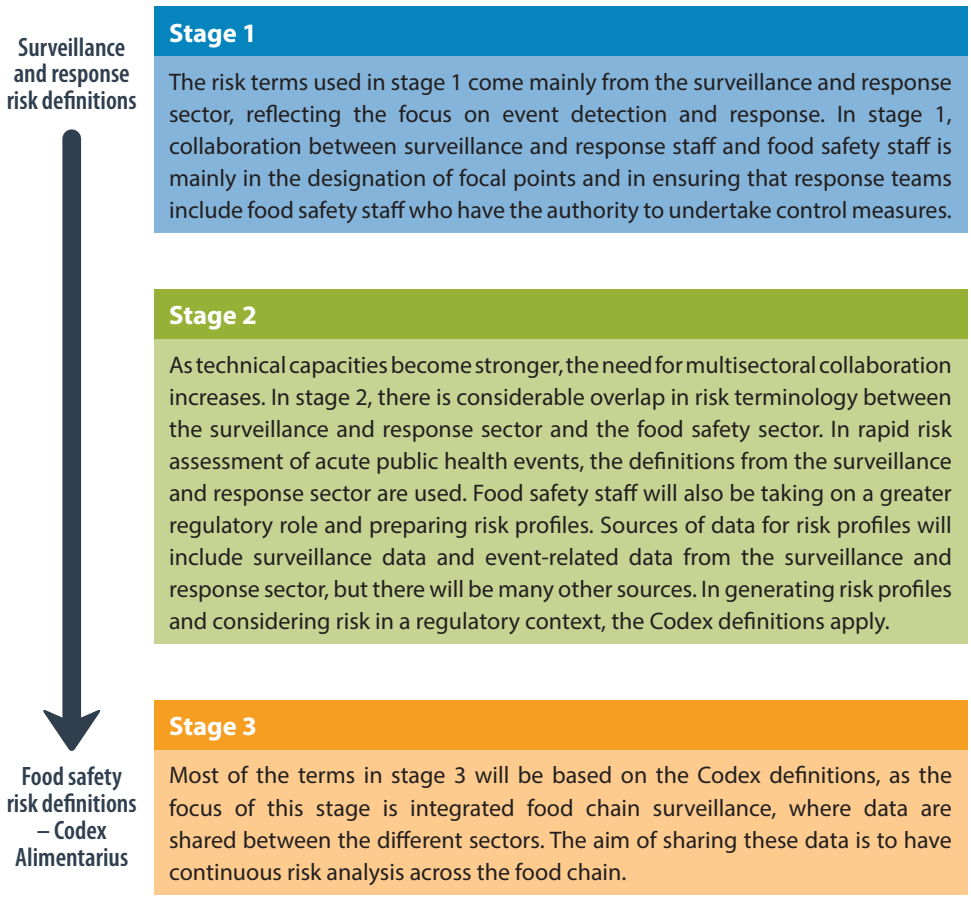


TABLE A2.1

Definitions of risk-related terms in the surveillance and response sector and the food safety sector

Term	Surveillance and response sector	Food safety sector
Context assessment	Assessment of the environment in which an event is taking place.	No definition
Exposure assessment	Evaluation of the potential exposure of individuals and populations to the hazards identified in a hazard assessment.	The qualitative or quantitative evaluation of the likely intake of biological, chemical, and physical agents via food, as well as exposures from other sources if relevant.
Hazard	An agent that has the potential to cause adverse health effects in exposed populations.	A biological, chemical or physical agent in, or condition of, food with the potential to cause an adverse health effect.
Hazard assessment	Identification of the hazard (or potential hazards) causing an event and of the associated adverse health effects.	No definition
Risk	The likelihood that an adverse event will occur during a specified period and the likely magnitude of the consequences.	A function of the probability of an adverse health effect and the severity of that effect, as a consequence of one or more hazards in food.
Risk analysis	No definition	A process consisting of three components: risk assessment, risk management and risk communication.
Risk assessment	A systematic process for gathering, assessing and documenting information in order to assign a level of risk. Risk assessment includes three components — hazard assessment, exposure assessment and context assessment.	A scientifically based process consisting of the following steps: (i) hazard identification, (ii) hazard characterization, (iii) exposure assessment, and (iv) risk characterization.
Risk characterization	A process of assigning a level of risk to an event based on assessment of the hazard, exposure and context. If no quantitative model or comparison with a guidance value (e.g. in food safety risk assessments) is available, the process is based on the expert opinion of the team.	The qualitative or quantitative estimation, including attendant uncertainties, of the probability of occurrence and severity of known or potential adverse health effects in a given population, based on hazard identification, hazard characterization and exposure assessment.

(TABLE A2.1 Cont.)

Term	Surveillance and response sector	Food safety sector
Risk communication	The range of communication principles and activities, and exchange of information, required throughout the preparedness, response and recovery phases of a serious public health event between responsible authorities, partner organizations and communities at risk to encourage informed decision-making, positive behaviour change and the maintenance of trust.	The interactive exchange of information and opinions throughout the risk analysis process concerning risk, risk-related factors and risk perceptions among risk assessors, risk managers, consumers, industry, the academic community and other interested parties, including the explanation of risk assessment findings and the basis of risk management decisions.
Risk management	The process of weighing policy options in the light of a risk assessment and, if required, selecting and implementing appropriate intervention options, including regulatory measures. With respect to acute public health events, risk management is the process by which appropriate actions are taken to manage and reduce the negative consequences of acute public health risks.	The process, distinct from risk assessment, of weighing policy alternatives, in consultation with all interested parties, considering risk assessment and other factors relevant for the health protection of consumers and for the promotion of fair trade practices, and, if needed, selecting appropriate prevention and control options.
Risk profile	No definition	The description of the food safety problem and its context.

References

1. Whole genome sequencing as a tool to strengthen foodborne disease surveillance and response: module 1: introductory module. Geneva: World Health Organization; 2023 (https://iris.who.int/handle/10665/373459) Licence: CC BY-NC-SA 3.0 IGO.

Annex 3

Country self-assessment

This self-assessment tool helps determine the capacities that exist in a country and identify areas where further development is required. The results can be used to tailor a workplan to meet the needs of the country, based on the existing surveillance and response system.

Instructions

How To



1. Convene a small group of three or four key people. The people completing the form will vary from country to country, but should always involve surveillance and response staff from the Ministry of Health. It would also be useful to include food safety staff in the process, if possible.
2. Allocate approximately half a day to complete the self-assessment.
3. Be honest in your evaluation of each indicator. There is no right or wrong answer. The self-assessment will be used to develop a 2–3 year workplan.
4. Mark a response for each indicator, even if it appears to be beyond the capacity in your country.
5. Some indicators are repeated in each stage, because it is important to ensure that the capacity exists, regardless of the stage a country might be in. Where you have already given a response for a particular indicator, you may subsequently put simply "as above", when it appears again.

Stage one assessment

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Strategic goal 1. An indicator-based surveillance system that can monitor trends of disease syndromes and identify outbreaks of foodborne diseases					
A surveillance system for notifiable diseases that collects syndromic data from the local level, and collates the data at the national level on a regular basis					
Inclusion in the surveillance system of diseases and syndromes that may indicate foodborne disease (e.g. diarrhoea)					
A database to store the surveillance data					
Capacity to analyse surveillance data on a regular basis (e.g. every week or every two weeks) to monitor trends and detect outbreaks					
Regular publication of surveillance bulletins, showing the trends in syndromic data that may indicate foodborne disease					
A protocol that documents the functioning of the surveillance system					
Strategic goal 2. An event-based surveillance system capable of detecting foodborne events					
A national focal point to receive reports about events					
An event report form to capture information about an event					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
An event database to store information about reported events					
Health care workers and sanitary or food inspectors have been trained on reporting foodborne events to EBS					
Strategic goal 3. Capacity to undertake rapid risk assessments of acute public health events such as foodborne disease outbreaks					
A team at the national level that can rapidly assess acute public health events					
A rapid risk assessment protocol adapted to deal with suspected foodborne disease events					
Capacity to assess all suspected foodborne events within 24 hours of the initial report					
Strategic goal 4. Outbreak response teams (at national level, as a minimum) can gather epidemiological evidence during foodborne outbreaks					
Appropriate people have been nominated at the national level to take part in outbreak response teams					
The people identified to take part in the outbreak response teams are trained to carry out investigations of foodborne diseases					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
During the response to an outbreak, the outbreak response team undertakes the following: • interviews people with the disease using a standard questionnaire; • develops and applies a case definition; • describes the number of cases using a line list; • conducts a descriptive analysis of the data by time, place and person;					
A response protocol, which documents each step the outbreak response team should take when investigating a suspected foodborne disease outbreak					
At least one epidemiologist in the country who can conduct analytical studies, where appropriate					
Foodborne disease outbreaks are summarized in outbreak reports					
Key information about each event and outbreak is logged in an event database					
The event database is regularly analysed and the results are published in the surveillance bulletin.					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Strategic goal 5. Outbreak response teams have the capacity to collect and transport appropriate specimens to a laboratory for identification of the etiological agent during foodborne outbreaks					
The people identified to take part in outbreak response teams have been trained to collect the appropriate clinical specimens during an outbreak investigation					
Clinical specimens are being regularly collected in foodborne disease outbreak investigations					
Sample collection and transportation are included in the response protocol					
There is an up-to-date list of laboratories that can perform the necessary testing					
If laboratory capacity does not exist in the country, referral pathways for specimens to be tested at regional laboratories have been documented					
Strategic goal 6. Multisectoral collaboration that facilitates rapid information exchange and support during foodborne disease outbreaks					
The surveillance and response staff know who the focal points are for food safety, animal health and the key laboratories that would be required to test clinical or food samples collected during an event					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
There is an effective (formal or informal) mechanism for rapid information exchange during suspected foodborne outbreak investigations between all the stakeholders and relevant sectors					
Strategic goal 7. **Monitoring and evaluating the surveillance and response system for foodborne diseases**					
Monitoring indicators for each component of the system (e.g. IBS, EBS, rapid risk assessment of acute foodborne events, response and multisectoral collaboration)					
A process for measuring the monitoring indicators (e.g. define when system will be monitored, how it will be monitored and by whom)					
A log of system performance					
Regular evaluation of the surveillance and response system in relation to foodborne diseases					

Stage two assessment

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Strategic goal 8. An indicator-based surveillance system that includes laboratory analysis, to allow better understanding of trends in foodborne diseases and to increase the sensitivity and specificity of outbreak detection					
A list of priority foodborne diseases for surveillance selected through a formal process					
Laboratory-based surveillance for priority foodborne diseases, in which cases detected through the surveillance system are confirmed and further characterized in the laboratory					
Protocols for collecting clinical specimens for all priority foodborne diseases that include: • objectives of the surveillance system, • which specimens will be collected (e.g. stool), • when specimens will be collected (e.g. every 20th patient meeting the case definition of diarrhoea), • how specimens will be collected, • how specimens will be stored before being transported to the laboratory, • where and how the specimens will be transported to.					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Protocols for testing clinical specimens for all priority foodborne diseases that includes: • a description of how laboratory testing is organized e.g. identifying what samples from which reporting sites go to which laboratories, • instructions for the further characterization of priority foodborne pathogens, • instructions for antimicrobial susceptibility testing of foodborne pathogens and how this links to the broader antimicrobial surveillance system.					
A database to house the laboratory-based surveillance data, with a data dictionary					
Data reporting protocols for all priority foodborne diseases that include: • who will send/ enter the data to the surveillance system, • what data will be sent, • how often the data will be sent, and					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
• what actions will be taken on the basis of the information sent to the surveillance system					
Antimicrobial susceptibility testing as a routine part of the surveillance system for relevant foodborne diseases					
Strategic goal 9. There is a fully functional notifiable disease surveillance system that can successfully monitor trends and detect outbreaks of foodborne diseases					
Existing laws and decrees governing the national notifiable disease surveillance system are up to date and include priority foodborne diseases					
There are case definitions for each of the notifiable foodborne diseases					
There are notification forms and a clear mechanism for reporting (e.g. fax number, telephone notification, web-based system)					
Laboratories and health care workers: • are aware of their obligations to report positive and any relevant or unusual test results to the surveillance system, • have specific forms for notification of cases,					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
• have clear instructions for reporting that fits within the existing surveillance system					
There is a notifiable disease surveillance system database that: • can record all the information required under the minimum data requirements, • allows data to be entered easily, • allows data to be extracted easily for analysis, • can be accessed at any time, • is relatively stable over time to enable trends to be monitored.					
To support the surveillance process there is a: • data dictionary, • surveillance system log, • disease-specific surveillance log					
Surveillance data from both the notifiable disease surveillance system and the laboratories are analysed and interpreted regularly					
Data analyses are included in a regular surveillance bulletin that is available to all stakeholders					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Surveillance protocols include the list of notifiable diseases with, for each disease: • a case definition, • the reason for surveillance, • a data analysis plan for monitoring trends and the thresholds used for cluster detection, • the public health action required (e.g. response triggered by one case or a cluster of cases).					
Strategic goal 10. **A fully functional event-based surveillance system capable of detecting foodborne events**					
There is a 24-hour telephone hotline, fax or email to receive reports at national level					
The sensitivity of EBS has been strengthened through training of people outside the health system (e.g. media, village leaders, etc.)					
There is active scanning of the media at national and international level for information about possible foodborne events					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Strategic goal 11. The capacity for rapid risk assessment of foodborne events is strengthened at the subnational level					
Staff at the subnational level have been designated responsibility for conducting rapid risk assessments					
These staff at the subnational level have been trained in rapid risk assessment and the training included examples of foodborne disease events that have occurred					
A mechanism is in place that allows the national level to provide technical support and advice to the subnational level, as required					
Laboratory data are routinely used in the rapid risk assessment of foodborne disease events.					
Strategic goal 12. Capacity exists at subnational level to carry out analytical epidemiological studies during foodborne disease outbreak investigations					
Specific questionnaires are available for each priority foodborne pathogen					
Capacity to conduct analytical epidemiological studies during outbreak investigations exists at the national and subnational levels					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Representatives from food safety and public health laboratories (and animal health, where applicable) are routinely part of the outbreak response team					
Strategic goal 13. Ad hoc or periodic research studies are conducted to gain a better understanding of foodborne diseases and their sources					
A mechanism exists for discussing, agreeing, planning and undertaking ad hoc research studies					
Strategic goal 14. Multisectoral collaboration facilitates the sharing of data for risk profiling					
There is a functioning communication mechanism between all stakeholders in food safety in the country; this requires agreement on: • what information is to be shared, • when information needs to be shared, • who needs to know the information, and • how information is to be shared					
There is evidence that the communication mechanism is in operation, e.g. a written communications plan, report from an outbreak debriefing about how the teams will work together better, etc.					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
There is multisectoral involvement in risk profiling of food safety problems, to help identify appropriate risk management strategies					
Strategic goal 15. **Monitoring and evaluation of the surveillance and response system for foodborne diseases**					
Adjust monitoring indicators for each component of the system to include: • laboratory-confirmed foodborne diseases for IBS; • increased sensitivity and specificity of EBS; • subnational capacity for rapid risk assessment and response; • increased multisectoral collaboration					
Regular evaluation of the surveillance and response system in relation to foodborne diseases					

Stage three assessment

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
Strategic goal 16. **Integrated food chain surveillance system that allows a better understanding of risks across the food chain**					
The existence of a team with members from each of the relevant sectors who are routinely sharing data on a regular basis					
A governance structure that allows data to be shared, and that includes a coordination and a communication mechanism					
The team can: • identify available data sources in each sector; • identify the appropriate pathogens for integrated food chain surveillance; • determine the animal species and foods to include					
A database that houses the integrated food chain surveillance data, with a data dictionary					
A data transfer mechanism extracts data from existing surveillance databases and other data sources to send to the integrated food chain surveillance database; the mechanism specifies:					

Indicator	Yes	Partially	No	Description of what is in place for this Indicator	Description of what is not in place for this Indicator
• the type of electronic transfer (e.g. automatic feed, manually sending spreadsheets); • the frequency of data transmission; • the data fields to be sent to the database A surveillance log is used to document changes in the integrated food chain surveillance system					
Multisectoral analysis and interpretation of the integrated data including: • a data quality review process; • source attribution					
Data analyses are included in a regular surveillance bulletin that is available to all stakeholders					
Outputs from the integrated food chain surveillance system are routinely used in risk analysis					
Performance of the integrated food chain surveillance system is monitored using indicators					
Regular evaluation of the integrated food chain surveillance system					

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